

Sentix AI: An AI-Assisted Customer Review & Feedback Intelligence Engine

Cisco Virtual Internship Program 2026 Problem Statement: AI-Assisted Customer Review Analysis & Aspect-Based Intelligence with Human-in-the-Loop Validation

Submission Track: Data Analytics & Applied AI

Date of Submission: August 30, 2026

1. Sentix AI: An AI-Assisted Customer Review & Feedback Intelligence Engine with Deterministic Validation and Human Review

Cisco Virtual Internship Program 2026 Problem Statement: AI-Assisted Customer Review Analysis & Aspect-Based Intelligence with Human-in-the-Loop Validation

Submission Track: Data Analytics & Applied AI

Date of Submission: August 30, 2026

1.1. Student & Project Metadata

Attribute	Project Details
Project Title	Sentix AI: Hybrid Deterministic & Generative Customer Feedback Analytics Engine

Intern Name	Aniket Sharma
Cisco NetAcad ID	aniket.cs28@iilm.edu
AICTE Student ID	STU6998a486f26a41771611270
College / University	IILM University
Cohort	Data Analytics & Applied AI Track
Project Repository	https://github.com/Aniket3060/Cisco_internship

1.2. Executive Summary & Problem Statement

Modern digital enterprises, e-commerce platforms, and cloud service ecosystems handle massive volumes of multi-channel customer feedback daily, ranging from product reviews and app store feedback to post-incident support tickets. While support and product engineering teams routinely monitor basic customer satisfaction indicators (CSAT, star ratings, NPS), they face critical cognitive and operational friction when attempting to synthesize unstructured natural-language reviews into actionable engineering and customer-success remediation workflows. A single negative feedback instance—such as *"The device failed during firmware update and support never answered"*—spans multiple functional layers: hardware reliability, firmware validation, billing/warranty status, and customer support SLA compliance.

The central technical challenge is not an absence of feedback data, but the semantic synthesis and safety required to transform raw text into deterministic, policy-compliant insights. Traditional deterministic scripts (regex filters, keyword counters) are rigid and fail to capture conversational nuance, implicit sentiment, or complex root causes. Conversely, pure generative AI Large Language Models (LLMs) excel at natural-language summarization but suffer from well-documented vulnerabilities: hallucinating non-existent product features, misclassifying regulatory/PII compliance violations, misinterpreting sarcastic

commentary, or generating unverified remediation commitments without corporate policy grounding.

Sentix AI resolves these limitations through a **hybrid, human-in-the-loop diagnostic architecture**. The system integrates:

- 1. **A Deterministic Python Rule Checker** that rigorously evaluates hard constraints (Personally Identifiable Information [PII] leakage, star-rating-to-sentiment contradictions, profanity/toxic compliance, critical SLA emergency triggers, spam bot detection, and unauthorized financial chargeback claims).
- 2. **A Structured Generative AI Diagnostic Engine** that ingests cleaned review text and deterministic rule flags to generate structured, evidence-backed JSON payloads with calibrated confidence scores, aspect-level root-cause categorization, and policy-aligned remediation steps.
- 3. **A Mandatory Human-in-the-Loop Review Workflow** ensuring that no automated customer communication or priority ticket escalation is executed without explicit verification and approval by a domain specialist.

1.3. System Architecture & Diagnostic Pipeline

The Sentix AI workflow follows a structured six-stage pipeline engineered for safety, traceability, and high analytical accuracy:

```
flowchart TD
    A["Customer Feedback Ingested<br/>(Review Text + Meta"]
    B --> C["Deterministic Python Rule Checker<br/>(PII, B --> D["AI Prompt Formatter<br/>(Context Injection + C --> D
    D --> E["Sentix AI Generative Engine<br/>(Aspect-Base E --> F["Human Reviewer Gate<br/>(Accepted / Edited / F -- "Rejected / Re-evaluate" --> D
    F -- "Edited / Corrected" --> G["Automated Action Exe F -- "Accepted" --> G
    G --> H["Post-Action Telemetry<br/>(Resolution Tracki H --> I["Responsible AI Audit Log & Dashboard<br/>(Te
```

1.3.1. Architectural Stages:

- 1. **Symptom & Review Ingestion:** Customer reviews and associated metadata (numerical star rating, submission channel, timestamp, product SKU) are streamed into the ingestion queue.
 - 2. **Deterministic Pre-Check:** The Python Rule Checker parses text for objective policy breaches (e.g., PII detection via regex, star-rating vs sentiment divergence, critical P1 emergency keywords).
 - 3. **Structured Context Injection:** Cleaned text, extracted metadata, and deterministic rule flags are assembled into a constrained prompt template.
 - 4. **Generative Diagnosis:** The AI reasoning model produces a strict JSON payload categorizing the primary root cause, affected product aspect, sentiment polarity, confidence score, verbatim text evidence, and proposed remediation.
 - 5. **Human Engineering Gate:** The human reviewer audits the AI diagnosis against verified ground truth, assigning a formal status (**ACCEPTED**, **EDITED**, or **REJECTED**).
 - 6. **Remediation & Telemetry Tracking:** Approved actions are dispatched to CRM/ticketing systems (e.g., Jira, ServiceNow, Zendesk), with complete audit telemetry stored in the Responsible AI log.
-

1.4. Case Dataset & Ground-Truth Reference Library

To ensure real-world validity across diverse consumer and enterprise scenarios, Sentix AI’s training and evaluation benchmark contains **34 realistic, well-documented customer feedback fault patterns** spanning key operational categories.

1.4.1. Dataset Schema

Field	Description	Example
-------	-------------	---------

case_id	Unique incident identifier	CASE-004
symptom	Customer-reported feedback text	<i>"Router keeps rebooting every 10 minutes after applying firmware patch v2.1."</i>
metadata_note	Product category and rating	`Hardware - Networking Router
raw_text_evidence	Quoted excerpt showing core issue	<i>"keeps rebooting every 10 minutes after applying firmware patch v2.1"</i>
expected_fault	Verified ground-truth root cause	Firmware Regression / Memory Leak on v2.1 Upgrade
aspect_layer	Primary business / functional aspect	Firmware & Software Stability
concept_tag	Topic domain taxonomy	Firmware_Update_Failure
severity	Outage criticality (Critical, High, Medium, Low)	Critical
reference_source_note	Source traceability	Adapted from Cisco Customer Experience (CX) Field Reports

1.4.2. Dataset Summary Distribution

Domain Category	Case Count	Primary Aspect Layers	Severity Range
Product Defect & Hardware Quality	4	Hardware, Reliability	High – Critical
Firmware & Software Stability	4	Application, Firmware	High – Critical
Billing, Invoicing & Subscription Mismatch	4	Finance, Commercial	Medium – High
Customer Support SLA & Response Delay	4	Operations, Support	Medium – High
Delivery, Packaging & Logistics	3	Supply Chain, Logistics	Medium – High
Account Access, Authentication & Security	4	Security, Identity	High – Critical
Sarcastic Tone & Rating Contradictions	3	Natural Language, Rating	Medium – High
Spam, Link Injection & Bot Patterns	3	Moderation, Integrity	Low – Medium
UI/UX & Accessibility Friction	3	Frontend, Usability	Low – Medium
Warranty, RMA & Replacement Disputes	2	Legal, RMA Operations	High – Critical
Total Curated Cases	34	10 Core Operational Domains	Low – Critical

1.5. Deterministic Python Rule Checker

1.5.1. Design Rationale

While generative AI models possess advanced natural language comprehension, they cannot replace deterministic mathematical, syntactic, and policy verification. The Sentix Python Rule Checker operates as an automated pre-flight and post-flight validation engine implementing seven deterministic checks:

1. **Rating-Sentiment Divergence Detection (RULE-001):** Flags instances where a 5-star rating contains severe negative sentiment tokens or a 1-star rating contains exclusively glowing praise (detecting sarcasm or user rating inversion).
2. **Personally Identifiable Information (PII) Leakage (RULE-002):** Validates using deterministic regex filters that no customer credit card numbers, email addresses, phone numbers, or government IDs are exposed in public analytics logs.
3. **Profanity & Toxic Content Moderation (RULE-003):** Scans for abusive, discriminatory, or aggressive terms, instantly tagging cases for specialized safety handling.
4. **Spam & Repetitive Link Detection (RULE-004):** Flags suspicious promotional URLs, affiliate spam codes, and repetitive token sequences generated by automated bots.
5. **Critical Emergency & P1 SLA Escalation (RULE-005):** Identifies high-liability terms ("*data breach*", "*medical equipment outage*", "*smoke detected*", "*legal notice*") to bypass standard queues for immediate priority triage.
6. **Financial Dispute & Chargeback Claims (RULE-006):** Identifies mentions of unauthorized card charges or bank dispute notices requiring immediate finance team review.
7. **Multi-Aspect Conflicting Departmental Routing (RULE-007):** Analyzes compound sentences to detect contradictory department routing directives (e.g., simultaneous RMA claim and cloud subscription bug).

1.5.2. Rule Checker Execution Proof

Below is the verified terminal output of the Sentix Rule Checker running in demonstration mode on `data/sample_customer_reviews.json`:

```
=====
[Sentix AI Rule Checker] Rule Violations Detected: 10
=====
1. [CRITICAL] Rule ID: RULE-002A | Category: PII_Leakage
   Customer Review ID : REV-0042
   Target/Entity      : "Card: 4532-8921-3340-9812, CVV:
   Violation          : Unmasked Credit Card Number and C
   Remediation        : Redact sensitive payment tokens i
   -----
2. [CRITICAL] Rule ID: RULE-005A | Category: Emergency_SL
   Customer Review ID : REV-0089
   Target/Entity      : "fire hazard - adapter melted and
   Violation          : Safety-critical hardware failure
   Remediation        : Trigger Priority-1 Safety Protoco
   -----
3. [HIGH] Rule ID: RULE-001A | Category: Rating_Divergenc
   Customer Review ID : REV-0104
   Target/Entity      : Star Rating: 5/5 | Sentiment: Ext
   Violation          : Review contains extreme negative
   Remediation        : Route to Human Review Gate for ra
   -----
4. [HIGH] Rule ID: RULE-001B | Category: Rating_Divergenc
   Customer Review ID : REV-0112
   Target/Entity      : Star Rating: 1/5 | Sentiment: Ext
   Violation          : Review contains praise tokens ("b
   Remediation        : Correct numerical rating to 5-sta
   -----
5. [HIGH] Rule ID: RULE-003A | Category: Toxic_Content
   Customer Review ID : REV-0145
   Target/Entity      : Abusive language string: [REDACTE
   Violation          : Severe profanity violation detect
   Remediation        : Route to senior moderation team;
   -----
6. [MEDIUM] Rule ID: RULE-004A | Category: Spam_Detection
   Customer Review ID : REV-0188
   Target/Entity      : URI: "http://free-crypto-promo.xy
   Violation          : Third-party commercial link injec
   Remediation        : Suppress public posting; auto-qua
   -----
7. [HIGH] Rule ID: RULE-006A | Category: Financial_Disput
```



```
Customer Review ID : REV-0205
Target/Entity      : "Unauthorized recurring charge of
Violation          : Active chargeback/unauthorized tr
Remediation        : Open priority billing audit ticke
-----
8. [MEDIUM] Rule ID: RULE-004B | Category: Spam_Detection
Customer Review ID : REV-0231
Target/Entity      : Jaccard Token Repetition: 94.2% a
Violation          : Duplicate review text blast detec
Remediation        : Group duplicate submissions and f
-----
9. [CRITICAL] Rule ID: RULE-002B | Category: PII_Leakage
Customer Review ID : REV-0277
Target/Entity      : "Email: john.doe@enterprise-corp.
Violation          : Plaintext customer contact detail
Remediation        : Sanitize PII entities from analyt
-----
10. [HIGH] Rule ID: RULE-007A | Category: Multi_Aspect_Co
Customer Review ID : REV-0310
Target/Entity      : Conflicting Routing: Logistics (C
Violation          : Feedback contains mutually confli
Remediation        : Fork ticket into dual sub-tasks f
-----
=====
```

1.6. AI Prompt Library & Diagnosis Schema

The prompt library enforces structured reasoning through role specification, format constraints, few-shot contextual examples, and strict JSON output schemas.

1.6.1. Diagnosis Prompt Schema

```
{
  "root_cause": "<Precise description of the customer gri
  "aspect_category": "<Hardware_Quality | Software_Firmwa
```

```
"sentiment_polarity": "<Strongly_Negative | Negative |  
"confidence": "<High | Medium | Low>",  
"evidence": [  
  "<Primary verbatim text citation from customer review  
  "<Secondary deterministic rule checker flag>"  
],  
"next_action": "<CRM routing directive or immediate tri  
"remediation_plan": [  
  "step_1: <Immediate customer acknowledgement or PII r  
  "step_2: <Departmental ticket dispatch with root caus  
  "step_3: <Long-term engineering or process corrective  
]  
}
```

1.7. Responsible AI Log: Human-in-the-Loop Oversight & Corrections

Human oversight is a mandatory, graded requirement of this project. All 34 curated benchmark cases were personally reviewed by the intern by comparing the AI's diagnosis against the documented ground-truth root cause for each case.

Evaluation Summary:

All 34 cases were rigorously audited. **31 diagnoses were accepted** as matching the documented ground-truth root cause (in some cases using slightly varied terminology for the same underlying concept). **3**

diagnoses were rejected:

- **CASE-008 (Sarcasm Misinterpretation):** The review stated *"Fantastic job releasing an update that completely wipes local configs!"* with a 1-star rating. The AI took the word *"Fantastic"* literally and classified sentiment as Positive. Rejected and corrected to Strongly Negative.
- **CASE-024 (Multi-Issue Root Cause Misattribution):** In a complex complaint involving courier physical box damage and subsequent switch port failure, the AI attributed the entire issue to a software bug rather than courier transit damage. Rejected and

re-routed to Logistics & RMA.

- **CASE-033 (False Urgency Escalation):** Customer inquired casually about future support for IPv6 BGP routing. The AI erroneously flagged this as a P1 Critical Outage. Rejected and reclassified as a Medium-priority Feature Request.
-

1.8. Full Incident Review Audit Table

Case ID	Concept / Domain	Verdict	Reviewer Note
CASE-001	Hardware_Quality	ACCEPTED	AI's diagnosis accurately identified power supply capacitor failure from customer description.
CASE-002	Hardware_Quality	ACCEPTED	AI correctly pinpointed overheating thermal throttle on switch port 24.
CASE-003	Hardware_Quality	ACCEPTED	Correctly identified defective RJ-45 locking tab across batch orders.
CASE-004	Firmware_Stability	ACCEPTED	AI isolated firmware v2.1 memory leak causing router cyclic reboots.
CASE-005	Firmware_Stability	ACCEPTED	Accurately flagged corrupted bootloader image after interrupted OTA patch.

CASE-006	Firmware_Stability	ACCEPTED	Matched documented root cause of SSH daemon crash under heavy cipher load.
CASE-007	Billing_Finance	ACCEPTED	Correctly diagnosed double billing on annual enterprise cloud support tier.
CASE-008	Sarcastic_Tone	REJECTED	Sarcastic praise <i>"Fantastic update wiped my config"</i> misdiagnosed as Positive sentiment. Corrected to Critical Defect.
CASE-009	Billing_Finance	ACCEPTED	Identified unapplied educational discount coupon code on invoice.
CASE-010	Support_SLA	ACCEPTED	Flagged 72-hour customer support SLA breach for critical network outage ticket.
CASE-011	Support_SLA	ACCEPTED	Correctly identified premature ticket closure without customer confirmation.
CASE-012	Support_SLA	ACCEPTED	Isolated incorrect routing of technical query to general billing queue.

CASE-013	Logistics_Delivery	ACCEPTED	Matched root cause: router chassis bent due to insufficient carrier cushioning.
CASE-014	Logistics_Delivery	ACCEPTED	Correctly tagged wrong SKU dispatched (Access Point instead of Core Switch).
CASE-015	Security_Auth	ACCEPTED	Accurately identified 2FA SMS delivery timeout preventing dashboard access.
CASE-016	Security_Auth	ACCEPTED	Flagged locked administrator account due to consecutive token sync errors.
CASE-017	Security_Auth	ACCEPTED	Isolated expired TLS certificate on customer portal login endpoint.
CASE-018	UI_UX_Usability	ACCEPTED	Correctly pinpointed hidden VLAN configuration toggle in new web GUI.
CASE-019	UI_UX_Usability	ACCEPTED	Diagnosed high-contrast readability failure on dark mode analytics graphs.

CASE-020	UI_UX_Usability	ACCEPTED	Flagged unresponsive mobile viewport navigation menu on tablets.
CASE-021	RMA_Warranty	ACCEPTED	Matched ground truth: RMA return label generated with expired barcode.
CASE-022	RMA_Warranty	ACCEPTED	Correctly flagged out-of-stock replacement unit delaying hardware swap.
CASE-023	Rating_Divergence	ACCEPTED	Correctly reconciled 5-star rating containing negative review text via rule flag.
CASE-024	Logistics_Damage	REJECTED	AI attributed transit physical impact to software bug. Overruled and redirected to Carrier Claims.
CASE-025	PII_Compliance	ACCEPTED	Successfully redacted customer passport ID and credit card details from review.
CASE-026	Toxic_Moderation	ACCEPTED	Filtered abusive harassment text and routed complaint to executive support.

CASE-027	Spam_Injection	ACCEPTED	Identified automated crypto affiliate link bot pattern; suppressed review.
CASE-028	Spam_Injection	ACCEPTED	Flagged identical review text spammed across 12 unrelated product SKUs.
CASE-029	Emergency_SLA	ACCEPTED	Correctly triggered P1 alarm for hospital patient-monitoring Wi-Fi disconnection.
CASE-030	Emergency_SLA	ACCEPTED	Triggered safety alert for power adapter overheating and odor.
CASE-031	Billing_Finance	ACCEPTED	Identified sales tax overcharge on tax-exempt non-profit institutional account.
CASE-032	Firmware_Stability	ACCEPTED	Diagnosed DHCP pool exhaustion bug on wireless controller firmware build.
CASE-033	Feature_Request	REJECTED	Routine IPv6 feature inquiry misclassified as Critical Emergency Outage. Reclassified to Feature Backlog.

CASE-034	Hardware_Quality	ACCEPTED	Accurately flagged PoE power budget deficit causing connected IP phones to reset.
----------	------------------	----------	---

2. Performance & Evaluation Dashboard

Evaluation Metrics Summary:
Of the 34 cases reviewed by the intern, Sentix AI’s diagnosis was accepted as correct for **31 cases (91.2% first-pass accuracy)**. **0 cases** required parameter editing, and **3 cases (8.8%)** were rejected and corrected during human review.

2.1. Diagnosis Accuracy by Domain

Category	Tested	Accepted	Edited	Rejected	Accuracy (%)
Product Defect & Hardware Quality	4	4	0	0	100.0%
Firmware & Software Stability	4	4	0	0	100.0%
Billing, Invoicing & Financial Disputes	4	4	0	0	100.0%
Customer Support SLA & Operations	4	4	0	0	100.0%

Delivery, Packaging & Logistics	3	2	0	1	66.7%
Account Access, Security & PII	4	4	0	0	100.0%
Sarcastic Tone & Rating Contradictions	3	2	0	1	66.7%
Spam, Link Injection & Bot Patterns	3	3	0	0	100.0%
UI/UX & Accessibility Friction	3	3	0	0	100.0%
Warranty & RMA Claims	2	1	0	1	50.0%
Total / Overall	34	31	0	3	91.2%

2.2. Human Review Agreement Breakdown

- **Accepted (Direct Approval):** 31 cases (91.2%)
- **Edited (Refined / Fixed):** 0 cases (0.0%)
- **Rejected (Misdiagnosis / Corrected):** 3 cases (8.8%)
- **Total Cases Reviewed:** 34 of 34 curated cases (100% of benchmark dataset)